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Transformations

Four moves, four sets of facts. “Describe fully” = the name of the move plus every fact it needs.
One move only.

THE TOPIC ON ONE MAP

Transformations

01 Translation

column vector $\begin{pmatrix} x \\ y \end{pmatrix}$

reverse: change both signs

02 Reflection

mirror line equation

$y = x$: swap

03 Rotation

angle

direction

centre

04 Enlargement

scale factor

centre

fractions

05 Combined

two moves → one

draw both

DESCRIBE FULLY The marks are in the facts — name the move, then give all of them:

ZONE TRANSLATION

1 fact

the column
vector

ZONE REFLECTION

1 fact

the mirror line
 $x = \dots$, $y = \dots$

ZONE ROTATION

3 facts

angle, direction,
centre

ZONE ENLARGEMENT

2 facts

scale factor,
centre

01 Translations

Every point slides the same way. Top number: right (+) or left (-). Bottom number: up (+) or down (-). The shape does not turn or change size.

WHEN YOU SEE → YOU DO

“translate by $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ ”

→ every vertex 4 left, 3 up

“describe” and the shape looks the same way round

→ translation: count from one corner to its image

“the vector that maps B back to A ”

→ change both signs

two translations one after the other

→ add the vectors

WORKED MODEL · JUNE 2024 P2HR

A has corner $(2, 2)$; the same corner of B is $(5, -3)$. Describe the transformation.

1 $5 - 2 = 3$ right, $-3 - 2 = -5$ down

same corner

ANSWER Translation $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$

× **TRAP** Writing coordinates $(3, -5)$ instead of a column vector. A translation needs $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$.

IDEA 01

Practise: Translations

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Translation

CH 5 • TOPIC 2 2 marks

Triangle with vertices $(1, 2)$, $(4, 2)$, $(1, 5)$ is translated by $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$. Write down the new vertices.

SELECTED PRACTICE

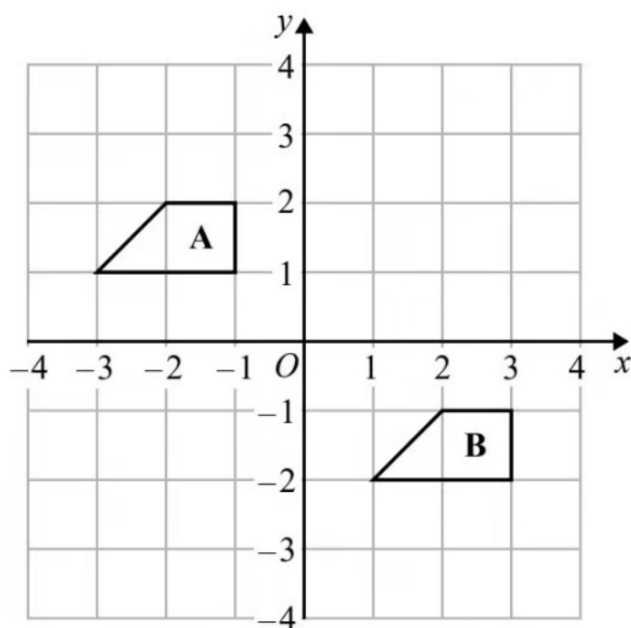
Q2

PRACTICE

Practice • Easy

CH 5 • TOPIC 2 2 marks

2



Describe the single transformation that maps shape **A** onto shape **B**.

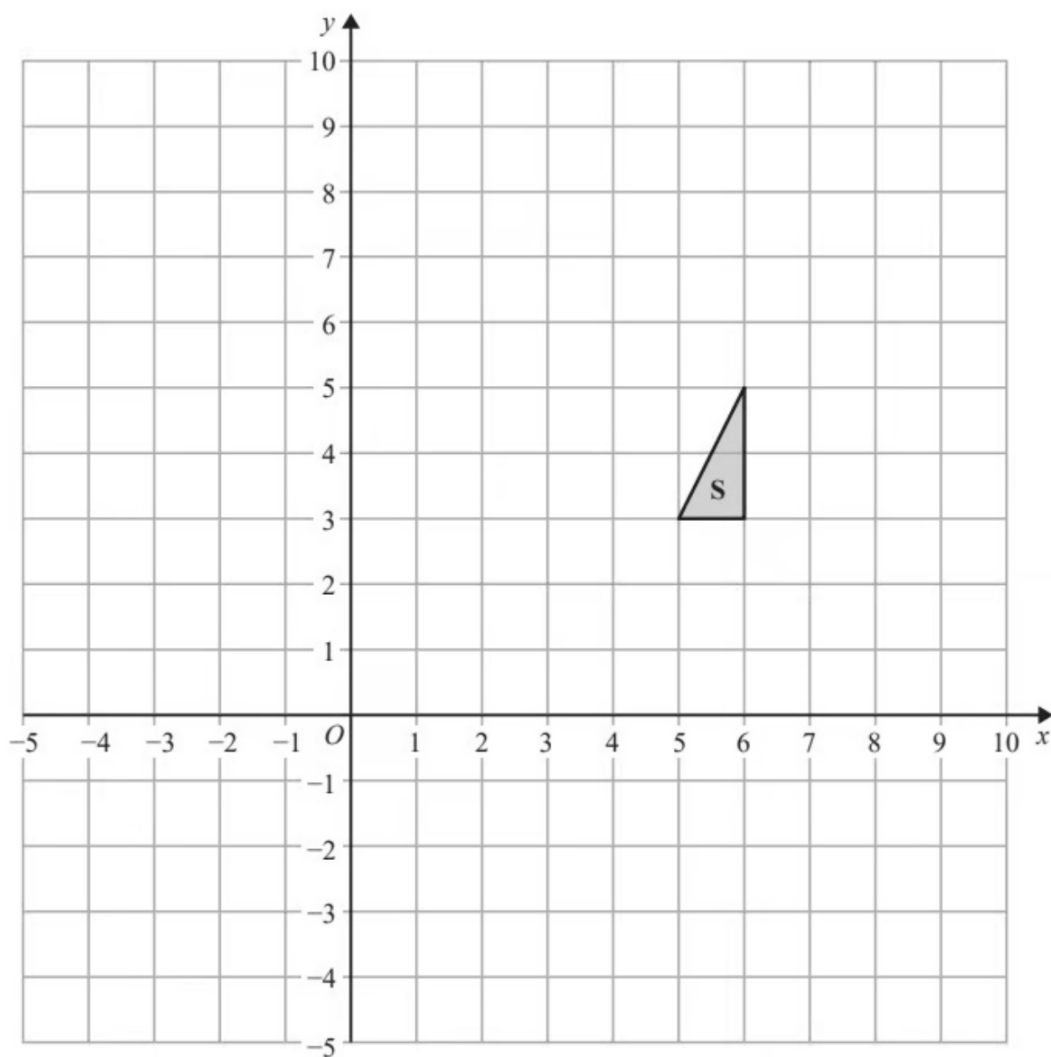
(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 5 • TOPIC 2 1 mark

10



Translate triangle **S** by the vector $\begin{pmatrix} -4 \\ -6 \end{pmatrix}$

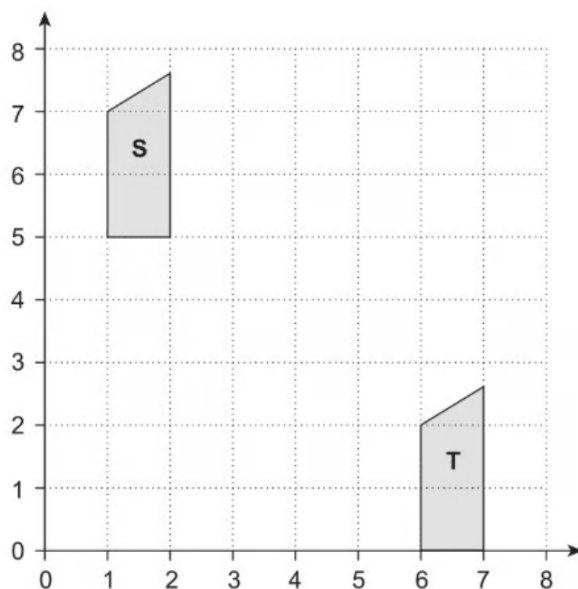
Label the new triangle **T**.

(1 mark)

Q4**PRACTICE**

Practice • Easy

CH 5 • TOPIC 2 2 marks

13 Here is a coordinate grid.

Shape S is translated to Shape T using vector $\begin{pmatrix} p \\ q \end{pmatrix}$

Write down the values of p and q .

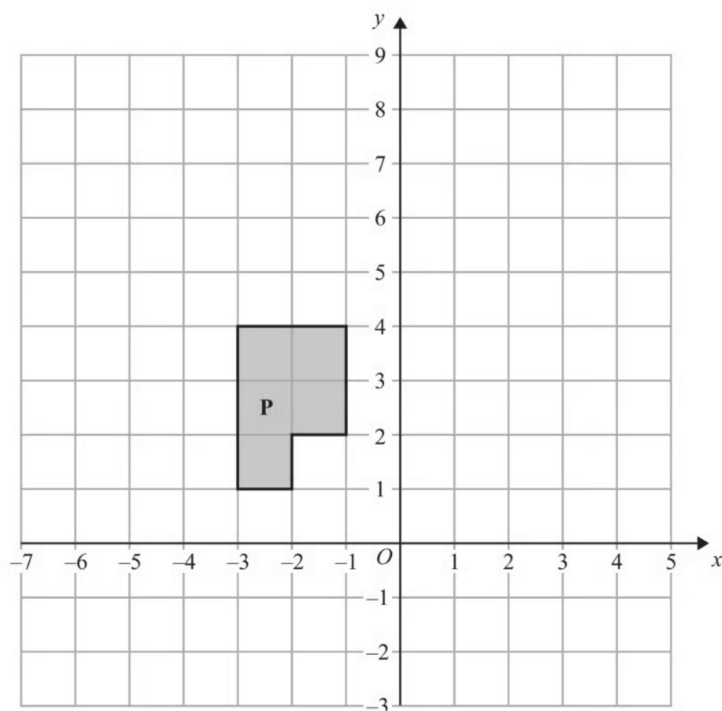
 $p = \dots\dots\dots$ $q = \dots\dots\dots$ **(2 marks)**

Q5

PRACTICE Practice • Medium

CH 5 • TOPIC 2 5 marks

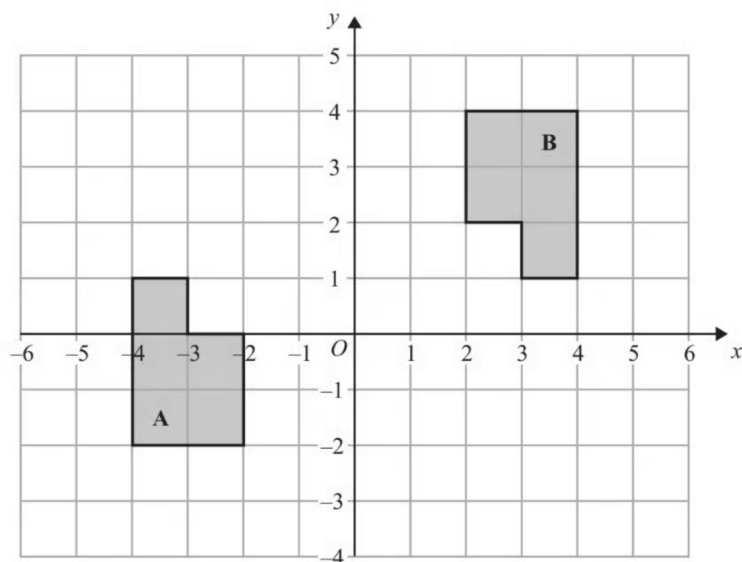
7 (a)



Translate shape **P** by the vector $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$

(2 marks)

(b)



Describe fully the single transformation that maps shape **A** onto shape **B**.

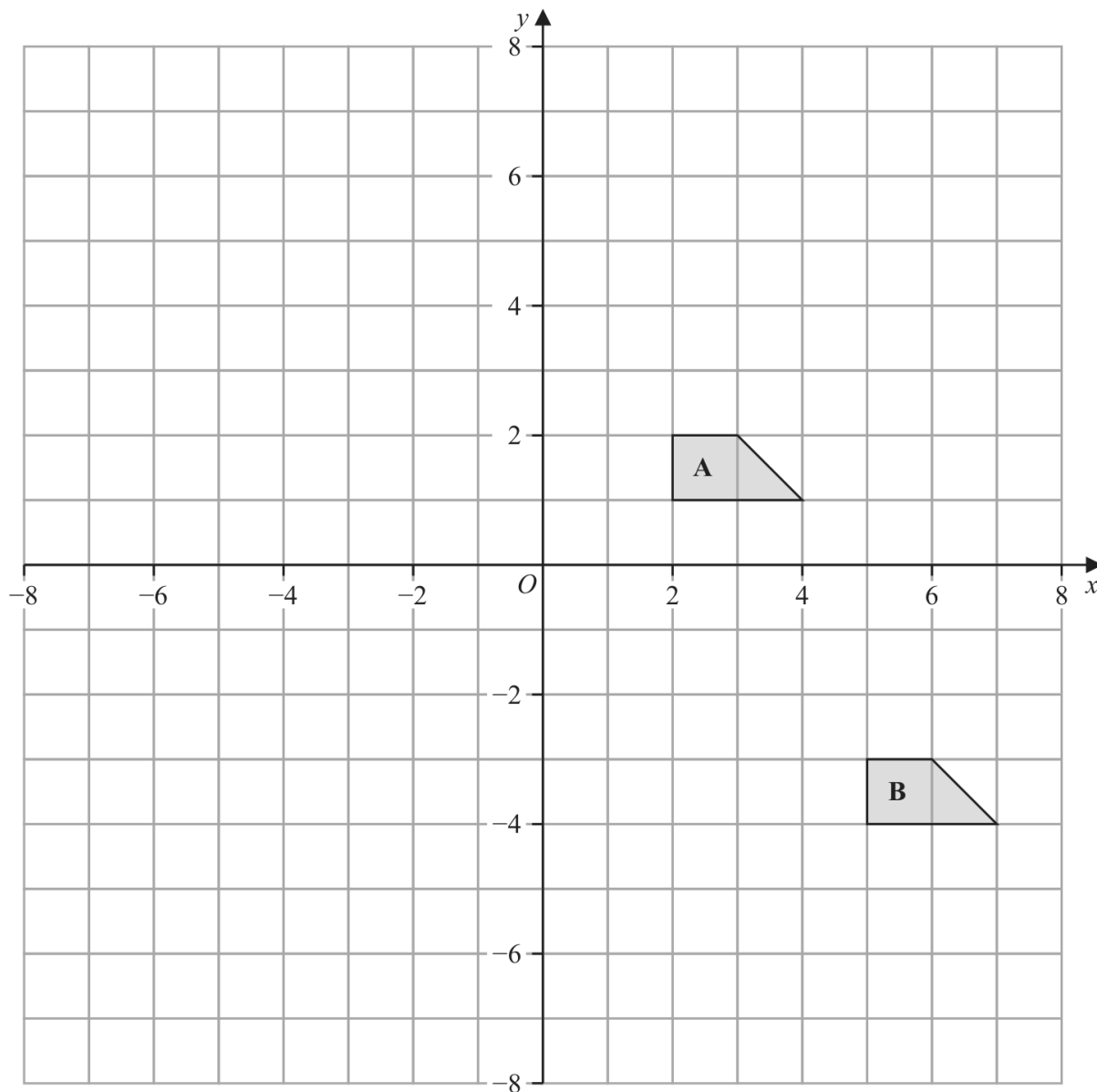
(3 marks)

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 • June 2024 • Paper 2HR • Q2

CH 5 • TOPIC 2 4 marks

2



- (a) Describe fully the single transformation that maps shape **A** onto shape **B**

- (b) On the grid above, rotate shape **A** 180° about $(-1, 0)$
Label your shape **C**

(Total for Question 2 is 4 marks)

02 Reflections

Each point goes straight across the mirror line and the same distance out the other side. Draw the line first, then do one vertex at a time.

WHEN YOU SEE → YOU DO

“reflect in $x = -1$ ”

→ vertical line: move across it horizontally

“reflect in $y = 2$ ”

→ horizontal line: move across it vertically

“reflect in $y = x$ ”

→ swap: $(a, b) \rightarrow (b, a)$

shape flipped, same size

→ reflection: find the line halfway between matching points

QUICK RULES x -axis: $(a, b) \rightarrow (a, -b)$ y -axis: $(a, b) \rightarrow (-a, b)$ $y = x$: $(a, b) \rightarrow (b, a)$
 $y = -x$: $(a, b) \rightarrow (-b, -a)$

WORKED MODEL • REFLECTION IN $x = -1$

Reflect $(2, 6)$ in the line $x = -1$.

1 $(2, 6)$ is 3 to the right of $x = -1$

distance to mirror

ANSWER 3 to the left: $(-4, 6)$

× **TRAP** Mixing up $x = 2$ and $y = 2$. $x = 2$ is a vertical line.

IDEA 02

Practise: Reflections

My questions first, then selected practice, then real past papers.

Q7**DR ESLAM**

Mirror line

CH 5 • TOPIC 2 **2 marks**

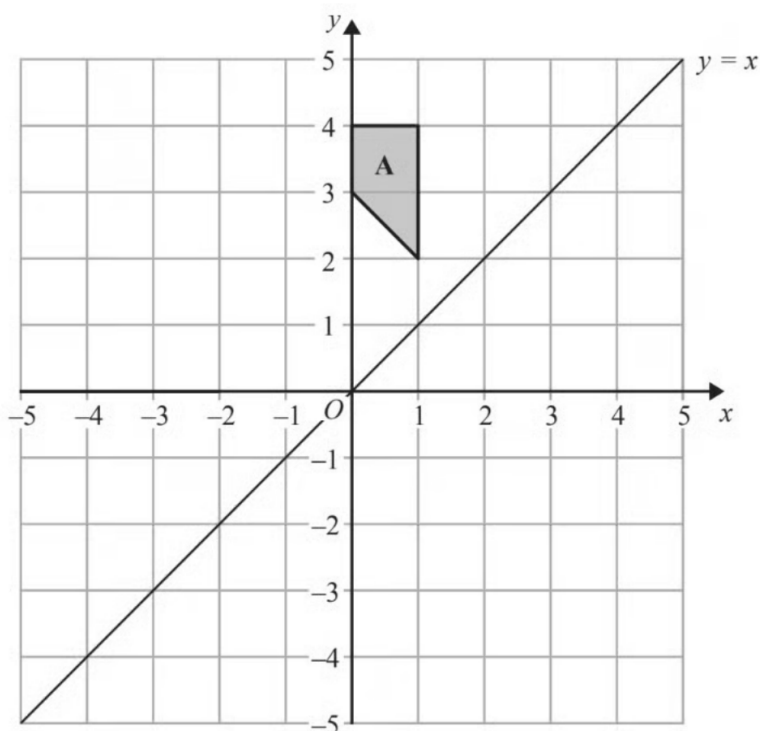
Reflect the point (5, 1) in (a) the line $x = 2$ (b) the line $y = x$ (c) the line $y = -1$.

SELECTED PRACTICE

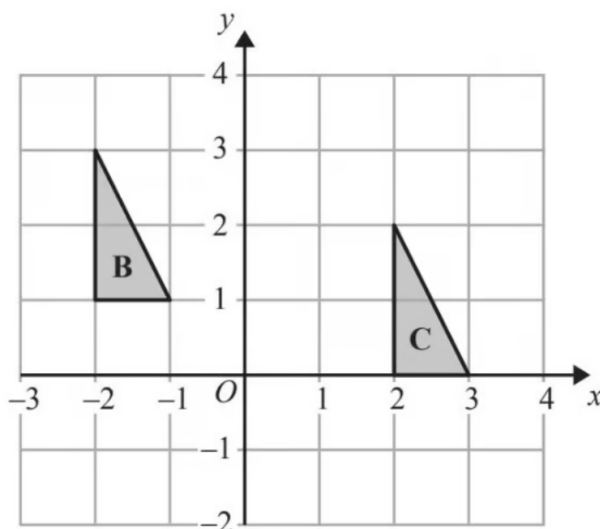
Q8**PRACTICE**

Practice • Easy

CH 5 • TOPIC 2 4 marks

5 (a)

On the grid, reflect shape **A** in the line $y = x$.

(2 marks)**(b)**

Describe fully the single transformation that maps triangle **B** onto triangle **C**.

(2 marks)

Q9

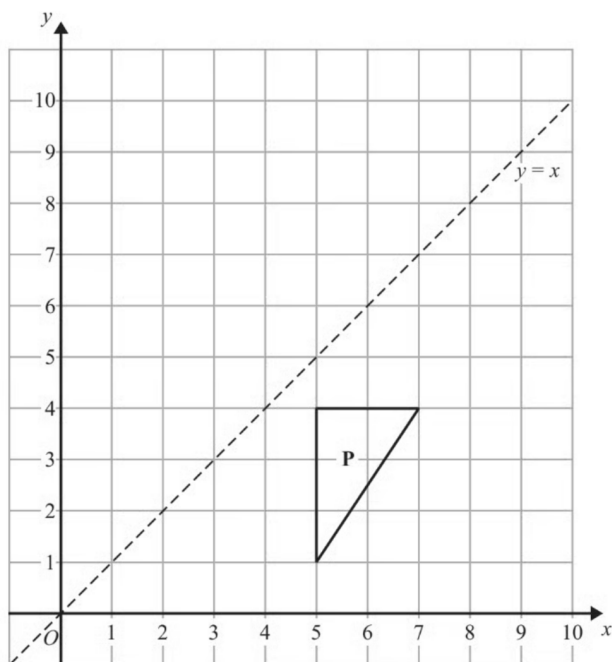
PRACTICE

Practice • Easy

CH 5 • TOPIC 2

4 marks

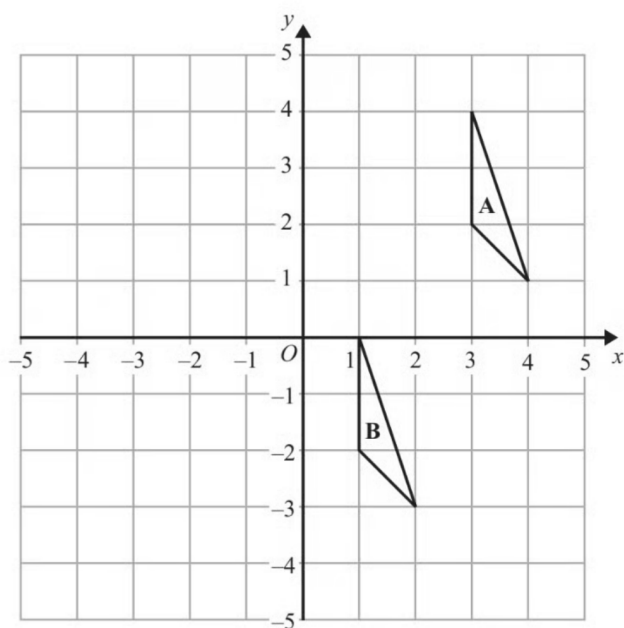
6 (a)



Reflect shape **P** in the line $y = x$.

(2 marks)

(b)



Describe fully the single transformation that maps triangle **A** onto triangle **B**.

(2 marks)

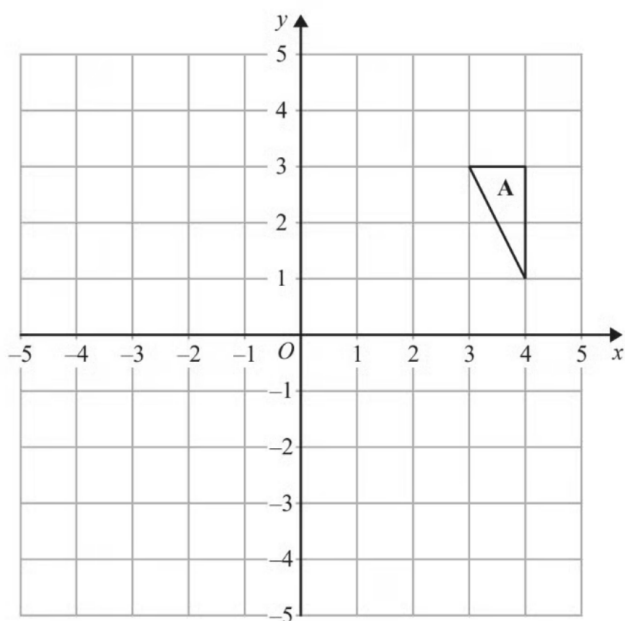
Q10

PRACTICE

Practice • Medium

CH 5 • TOPIC 2 4 marks

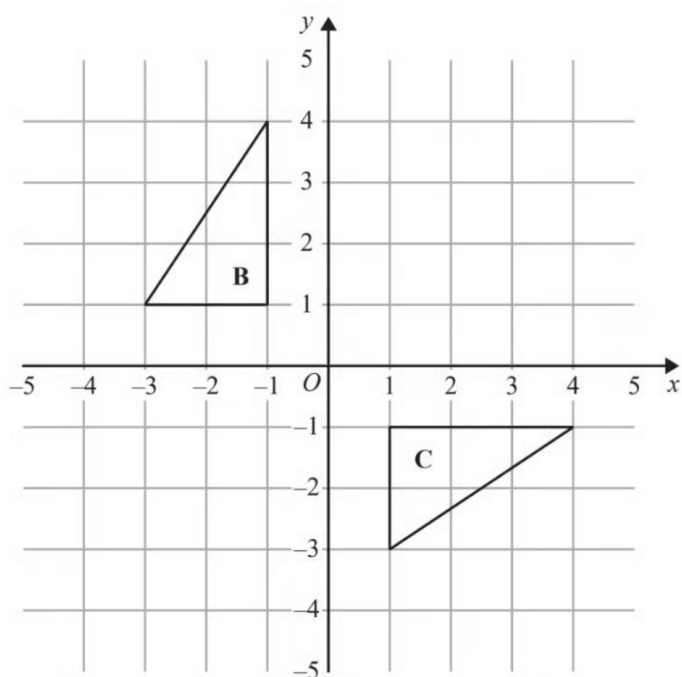
4 (a)



Rotate triangle **A** 90° anticlockwise with centre **O**.

(2 marks)

(b)



Describe fully the single transformation that maps triangle **B** onto triangle **C**.

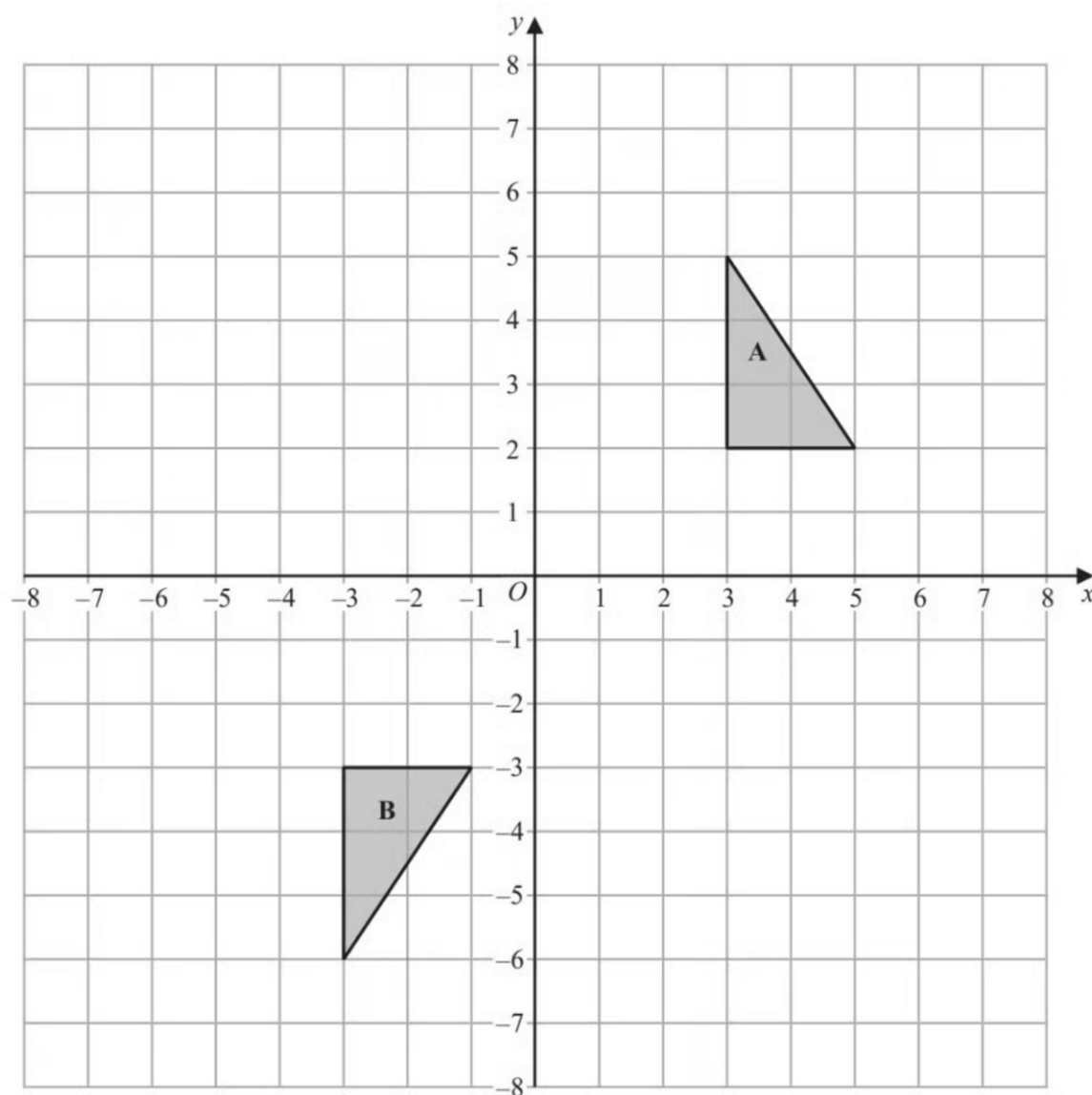
(2 marks)

Q11**PRACTICE**

Practice • Medium

CH 5 • TOPIC 2 3 marks

11



Shape **A** can be transformed to shape **B** by a reflection in the x-axis followed by a

translation $\begin{pmatrix} c \\ d \end{pmatrix}$

Find the value of c and the value of d .

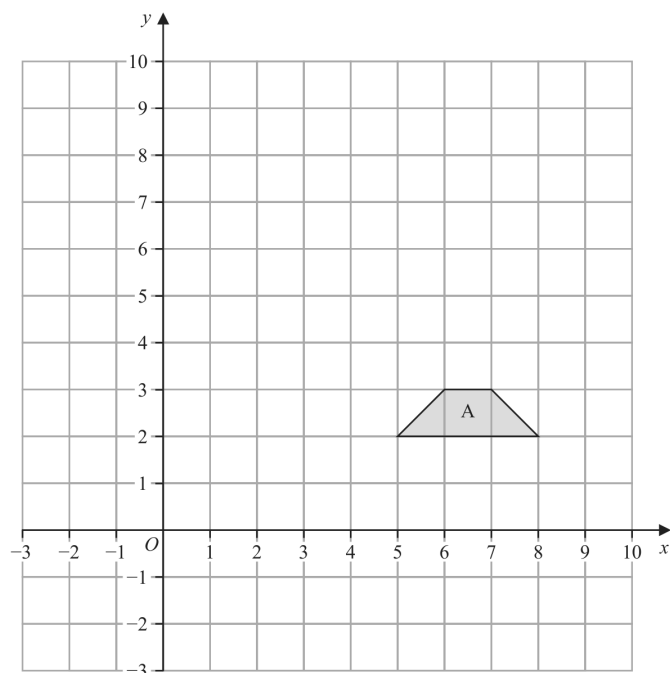
(3 marks)

PAST PAPERS

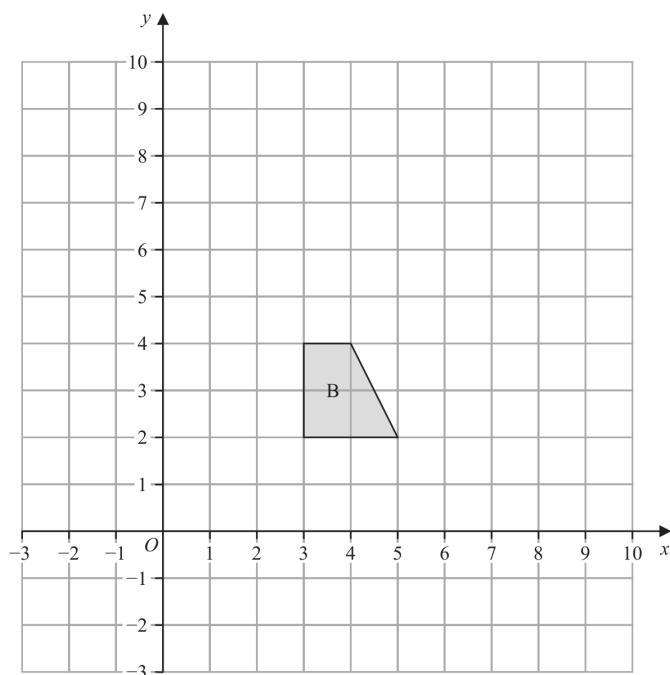
Q12 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 2HR • Q5

CH 5 • TOPIC 2 4 marks

5



(a) On the grid above, reflect shape **A** in the line $y = x$



(b) On the grid above, enlarge shape **B** by scale factor 2 with centre (1, 1)

(Total for Question 5 is 4 marks)

03 Rotations

Turn the shape about a fixed centre. Use tracing paper: pin the centre, turn the paper. To describe, give all three facts.

WHEN YOU SEE → YOU DO

“rotate 90° clockwise about (0, 1)”

→ trace, pin (0, 1), turn a quarter right

shape upside-down, same size

→ rotation 180°: centre is the midpoint of matching points

about the origin

→ 90° cw: $(a, b) \rightarrow (b, -a)$; 180°: $(a, b) \rightarrow (-a, -b)$

“describe fully”

→ angle + direction + centre

Pin the centre

Turn the tracing

Mark each vertex

Check one side's direction

WORKED MODEL • JUNE 2025 P2H

P has vertices (2, 2), (2, 6), (4, 2); Q has (8, 8), (8, 4), (6, 8). Describe the transformation.

1 Q is P turned upside-down: 180°

same size, flipped both ways

2 centre = midpoint of (2, 2) and (8, 8)

halfway

ANSWER Rotation 180° about (5, 5)

× **TRAP** Forgetting the direction for 90°. (For 180° no direction is needed.)

IDEA 03

Practise: Rotations

My questions first, then selected practice, then real past papers.

Q13 DR ESAM Rotation

CH 5 • TOPIC 2 2 marks

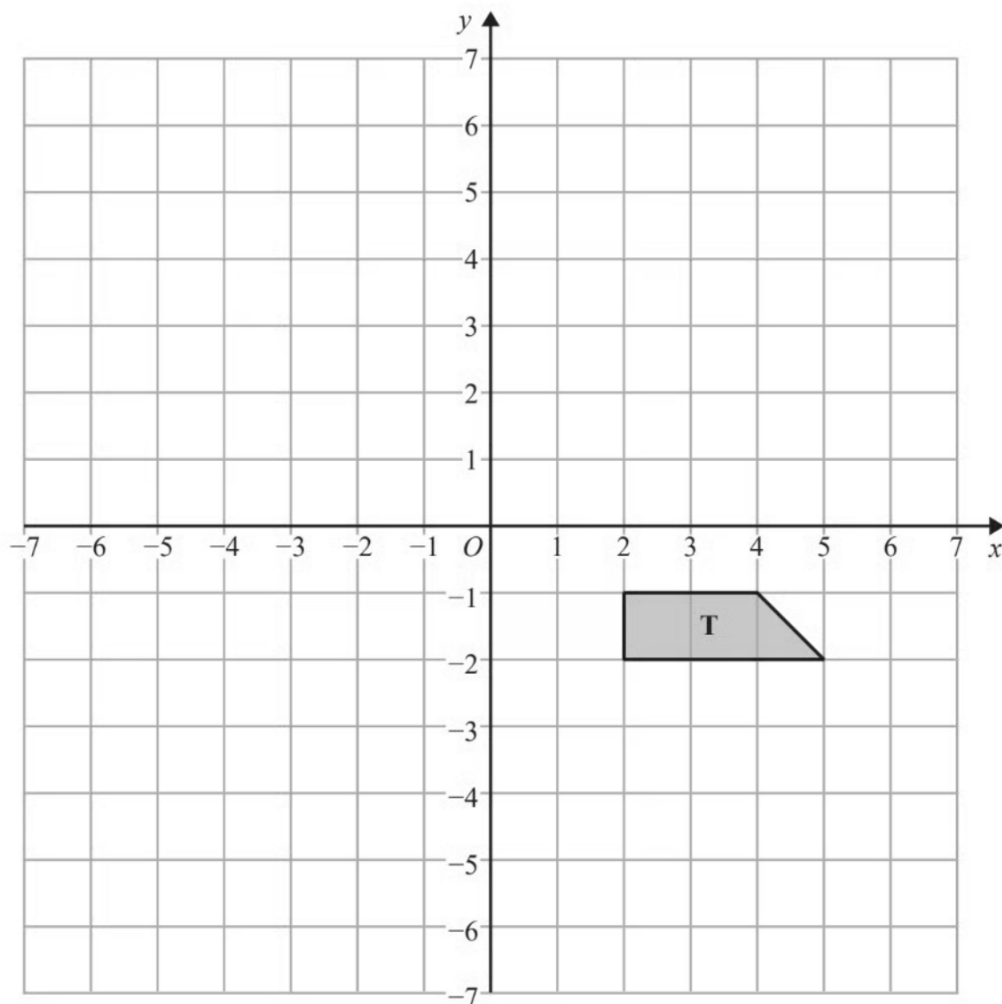
Rotate the point $(3, 1)$ through 90° clockwise about the origin, then 180° about the origin.

SELECTED PRACTICE

Q14 PRACTICE Practice • Easy

CH 5 • TOPIC 2 2 marks

1 (a)



Rotate trapezium **T** 180° about the origin.
Label the new trapezium **A**.

(1 mark)

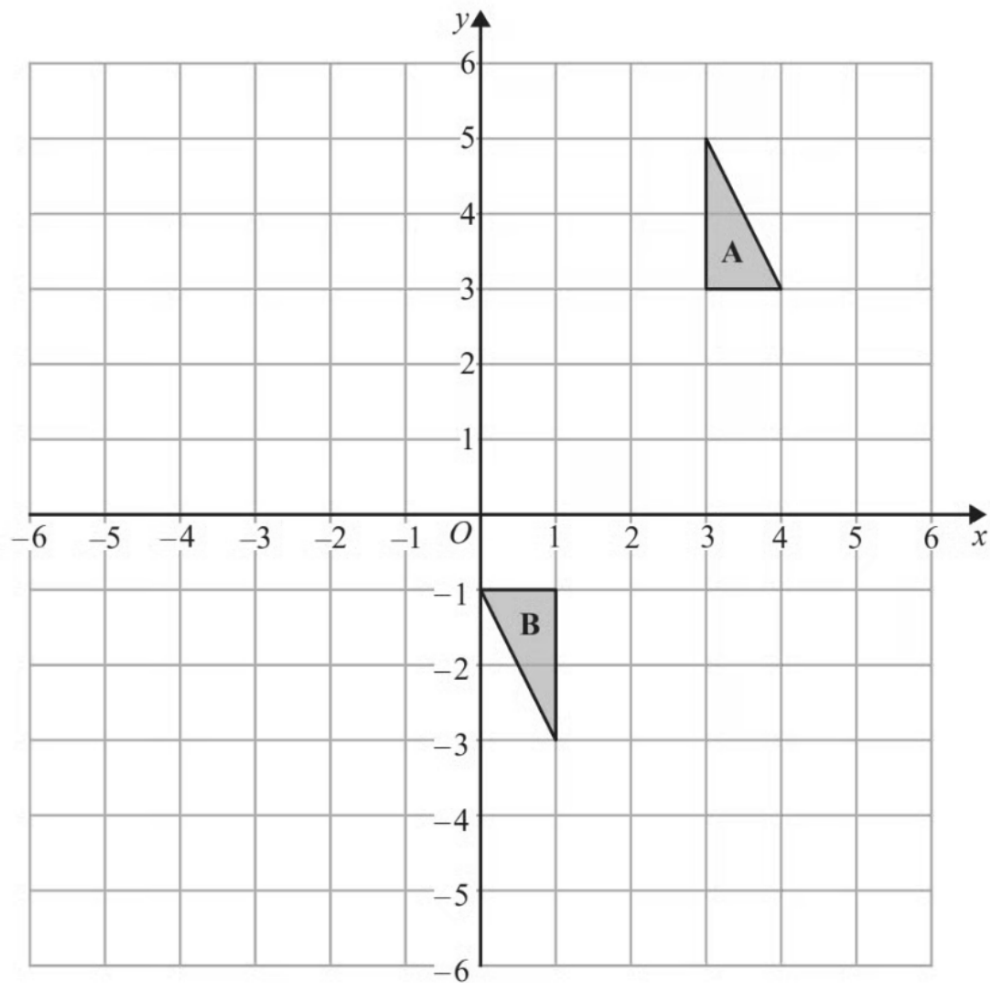
(b) Translate trapezium **T** by the vector $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$

Label the new trapezium **B**.

(1 mark)

Q15**PRACTICE**

Practice · Medium

CH 5 · TOPIC 2 **3 marks****2**

Describe fully the single transformation that maps triangle **A** onto triangle **B**.

(3 marks)

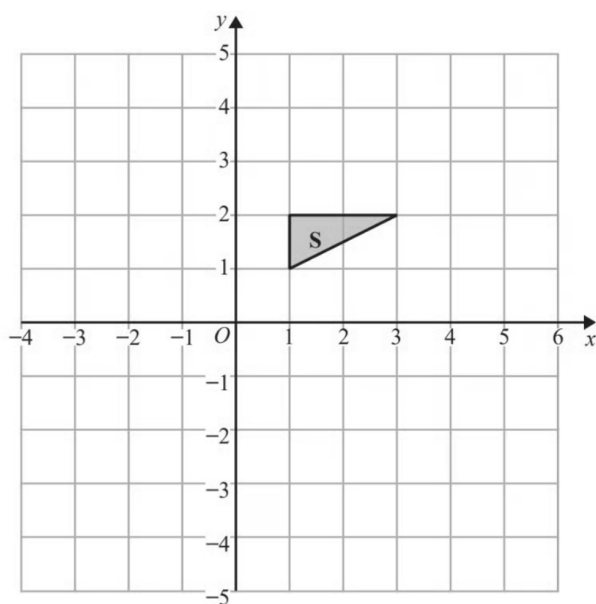
Q16

PRACTICE

Practice • Medium

CH 5 • TOPIC 2 5 marks

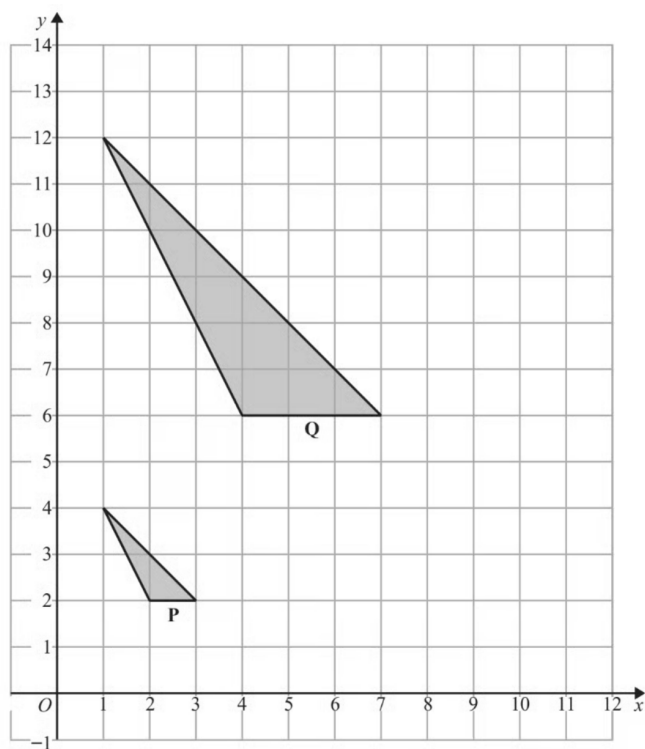
8 (a)



On the grid, rotate shape **S** by 90° anticlockwise about the origin.

(2 marks)

(b)



Describe fully the single transformation that maps shape **P** onto shape **Q**.

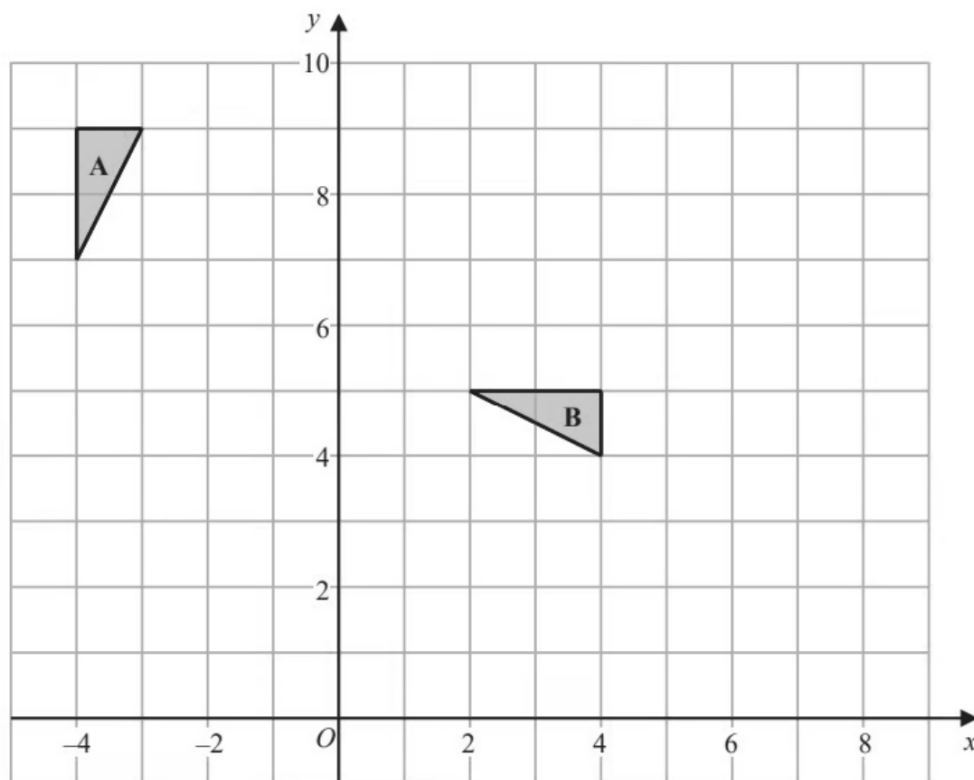
(3 marks)

Q17**PRACTICE**

Practice • Hard

CH 5 • TOPIC 2 3 marks

10



Describe fully the single transformation that maps triangle **A** onto triangle **B**.

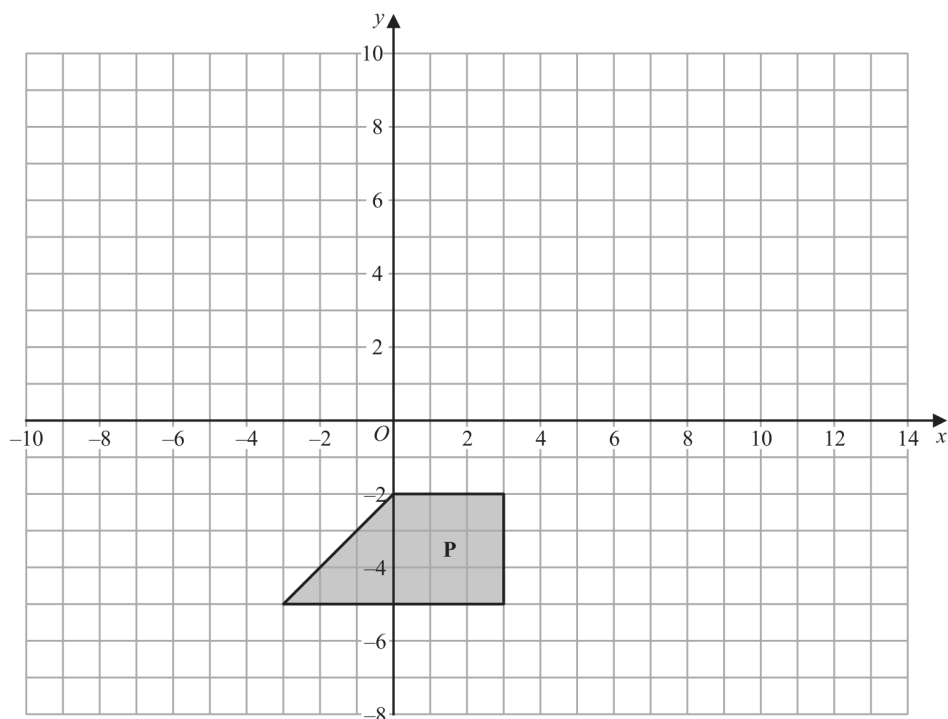
(3 marks)

PAST PAPERS

Q18 PAST PAPER Edexcel 4MA1 • January 2021 • Paper 2HR • Q5

CH 5 • TOPIC 2 6 marks

5 Here is a shape **P** drawn on a grid of squares.

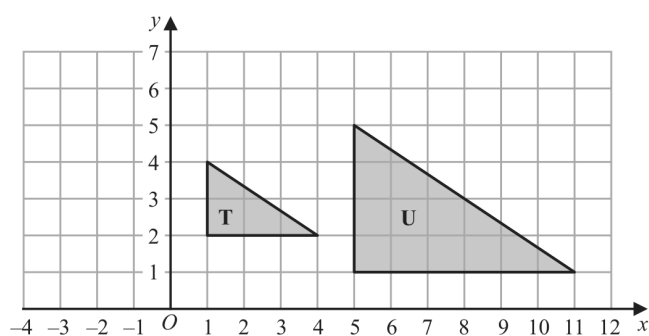


- (a) On the grid, rotate shape **P** 180° about the point $(-3, 2)$
Label the new shape **Q**.

(2)

- (b) On the grid, translate shape **P** by the vector $\begin{pmatrix} 10 \\ 8 \end{pmatrix}$
Label the new shape **R**.

Here are triangle **T** and triangle **U** drawn on a grid of squares.



- (c) Describe fully the single transformation that maps triangle **T** onto triangle **U**.

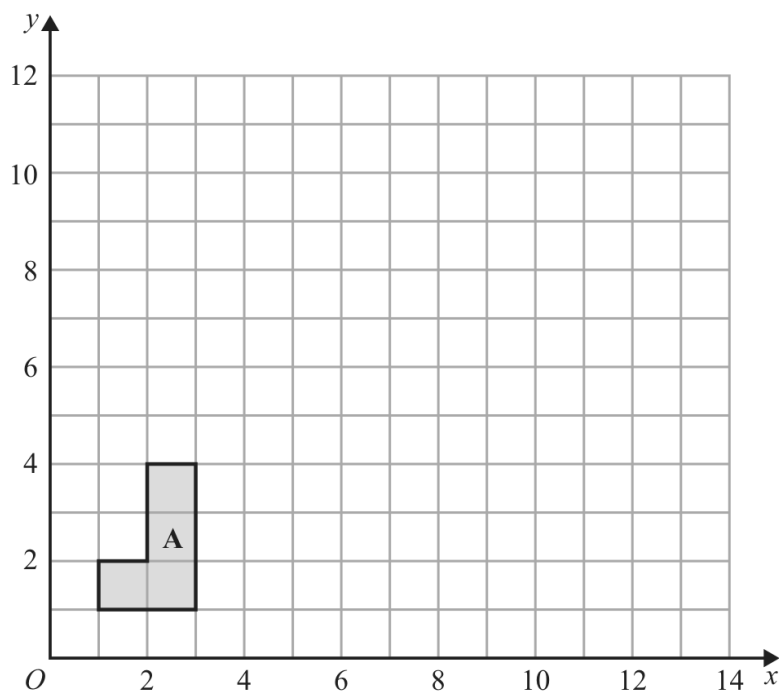
(Total for Question 5 is 6 marks)

Q19**PAST PAPER**

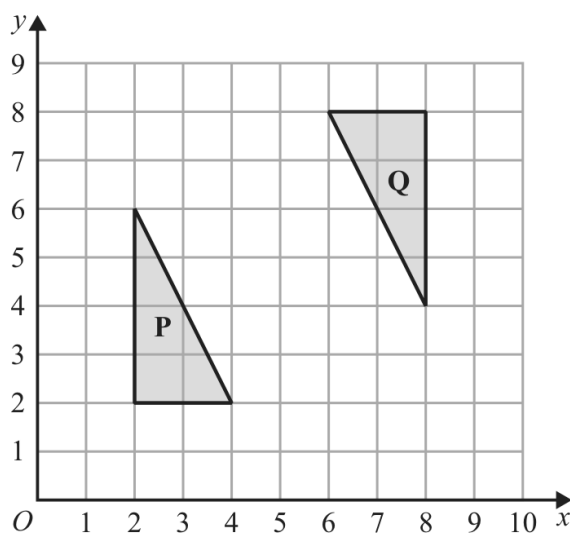
Edexcel 4MA1 • June 2025 • Paper 2H • Q2

CH 5 • TOPIC 2 5 marks

2



- (a) On the grid, enlarge shape A with scale factor 3 and centre (0, 1)



- (b) Describe fully the single transformation that maps triangle P onto triangle Q

(Total for Question 2 is 5 marks)

04 Enlargements

From the centre, every vertex goes scale-factor times as far. Scale factor < 1 : the shape gets smaller and moves closer. To describe: scale factor and centre.

WHEN YOU SEE → YOU DO

“scale factor 3, centre (1, 2)”

→ distance from centre $\times 3$ for every vertex

“scale factor $\frac{1}{2}$ ”

→ halve the distances: smaller, nearer the centre

same shape, different size

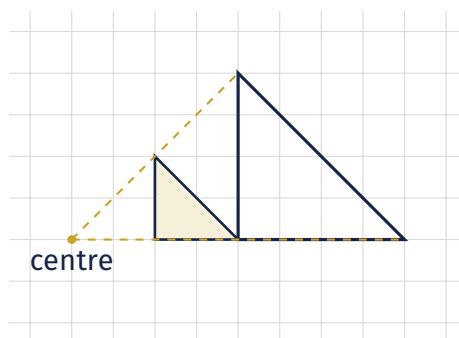
→ enlargement: join matching corners — lines meet at the centre

“area of the enlarged shape”

→ area $\times k^2$

a negative scale factor

→ go through the centre to the other side



centre to vertex: 2 right, 2 up
scale factor 2: 4 right, 4 up
lengths $\times 2$, area $\times 4$

WORKED MODEL • JANUARY 2020 P1H

Enlarge (7, 10), (15, 10), (15, 4), (11, 4) with scale factor $\frac{1}{2}$, centre (1, 2).

1 (7, 10): from centre $\begin{pmatrix} 6 \\ 8 \end{pmatrix} \rightarrow \begin{pmatrix} 3 \\ 4 \end{pmatrix} \rightarrow (4, 6)$

halve the vector

ANSWER (4, 6), (8, 6), (8, 3), (6, 3)

× **TRAP** Scale factor from the centre, not from a corner of the shape. Always measure from the centre.

IDEA 04

Practise: Enlargements

My questions first, then selected practice, then real past papers.

Q20 **DR ESLAM** Enlargement

CH 5 • TOPIC 2 3 marks

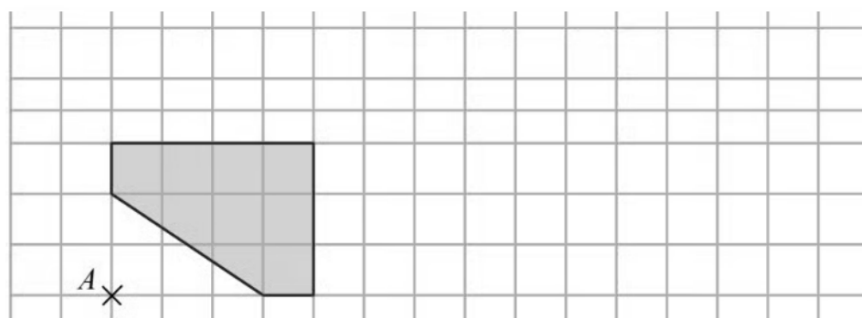
Triangle (2, 2), (4, 2), (2, 6) is enlarged with centre (0, 0) and scale factor $-\frac{1}{2}$. Write down the new vertices.

SELECTED PRACTICE

Q21 **PRACTICE** Practice • Easy

CH 5 • TOPIC 2 3 marks

7

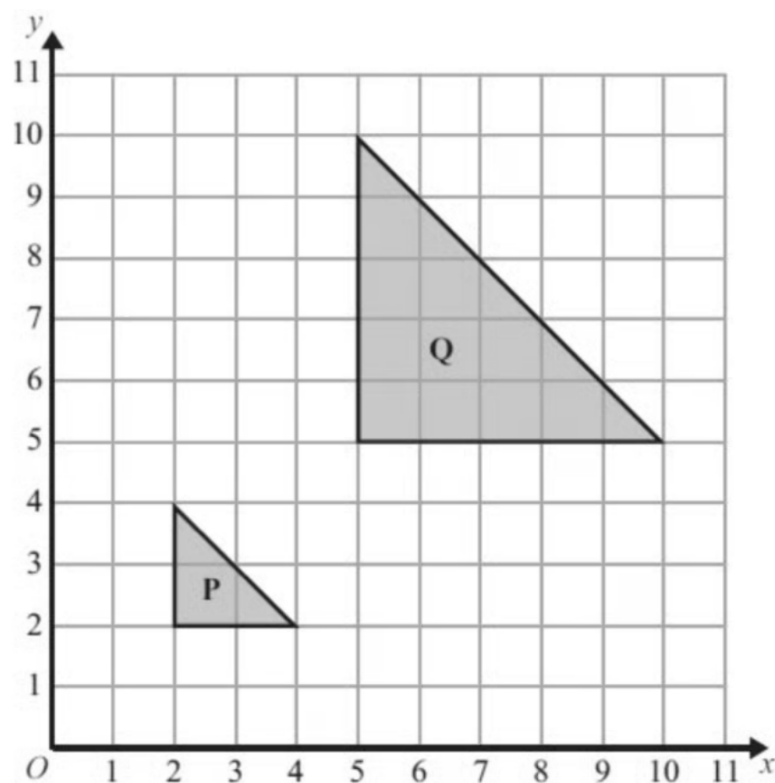


On the grid, enlarge the shape by scale factor 3, centre A.

(3 marks)

Q22**PRACTICE**

Practice • Medium

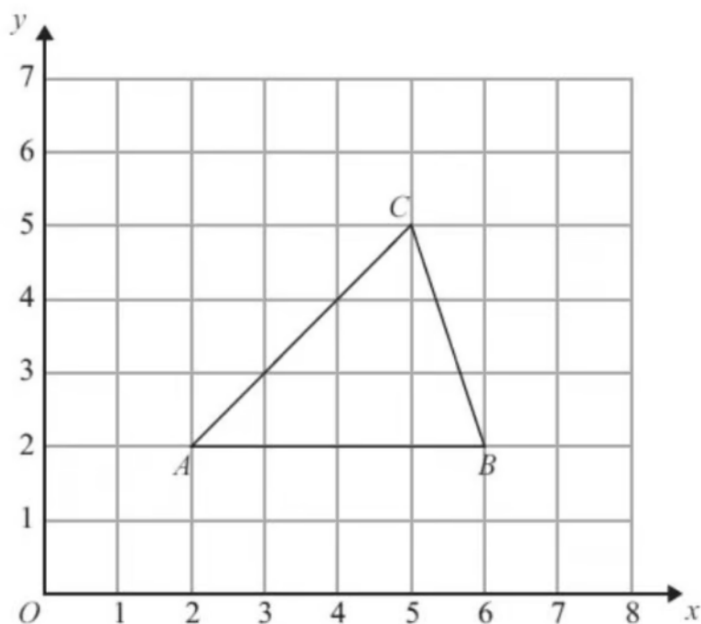
CH 5 • TOPIC 2 **3 marks****10**

Describe fully the single transformation that maps shape **P** onto shape **Q**.

(3 marks)

Q23**PRACTICE**

Practice • Hard

CH 5 • TOPIC 2 **3 marks****5**

Triangle ABC is drawn on a centimetre grid.

A is the point (2, 2).

B is the point (6, 2).

C is the point (5, 5).

Triangle PQR is an enlargement of triangle ABC with scale factor $\frac{1}{2}$ and centre (0, 0).

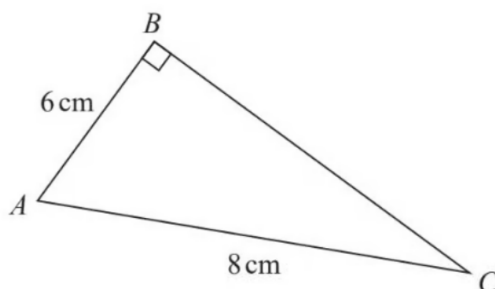
Work out the area of triangle PQR .

(3 marks)

Q24 PRACTICE Practice · Hard

CH 5 · TOPIC 2 2 marks

7 (a) ABC is a right-angled triangle.

Here is Sarah's method to find the length of BC .

$$BC^2 = AB^2 + AC^2$$

$$= 6^2 + 8^2$$

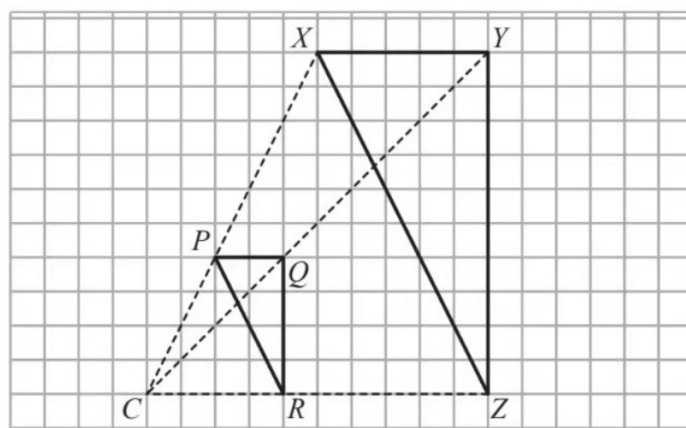
$$= 100$$

$$BC = 10$$

What mistake has Sarah made in her method?

(1 mark)

(b)

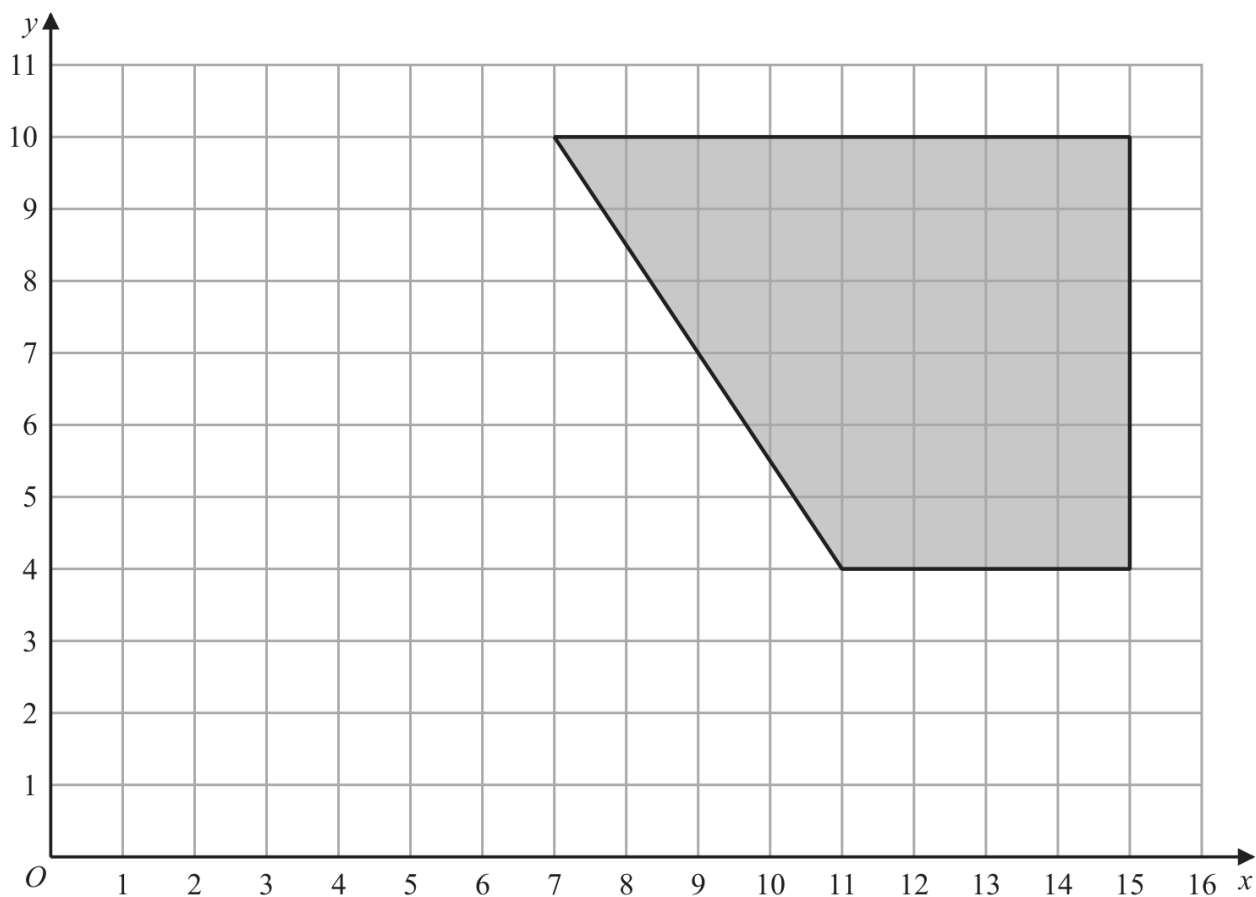
Roy is going to enlarge triangle PQR with centre C and scale factor $1\frac{1}{2}$ He draws triangle XYZ .Explain why Roy's diagram is **not** correct.

(1 mark)

PAST PAPERS

Q25**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q8

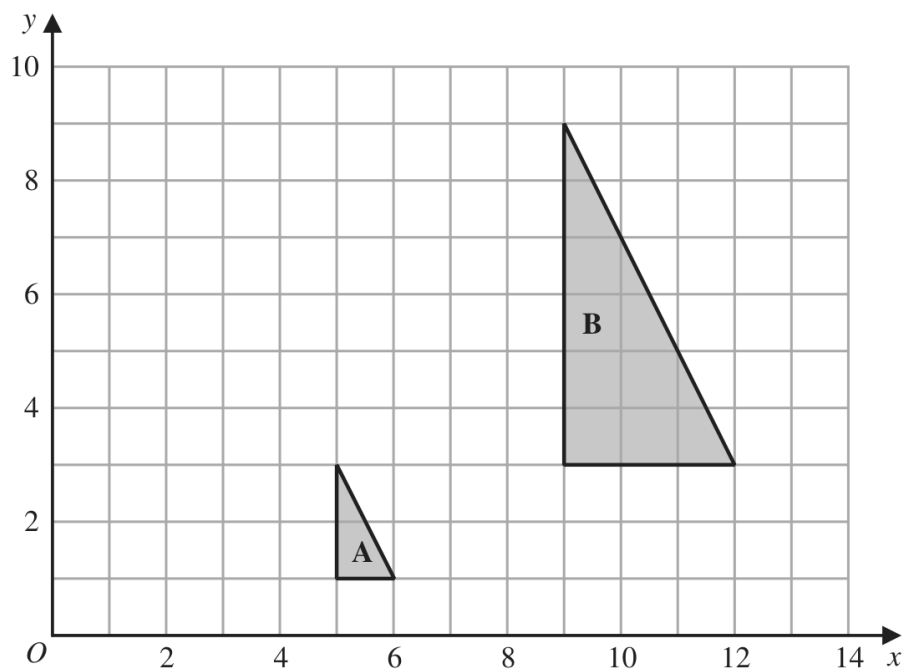
CH 5 • TOPIC 2 2 marks**8**

On the grid, enlarge the shaded shape with scale factor $\frac{1}{2}$ and centre (1, 2)

(Total for Question 8 is 2 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q1

CH 5 • TOPIC 2 4 marks**Answer ALL TWENTY FIVE questions.****Write your answers in the spaces provided.****You must write down all the stages in your working.****1**

(a) Describe fully the single transformation that maps triangle A onto triangle B

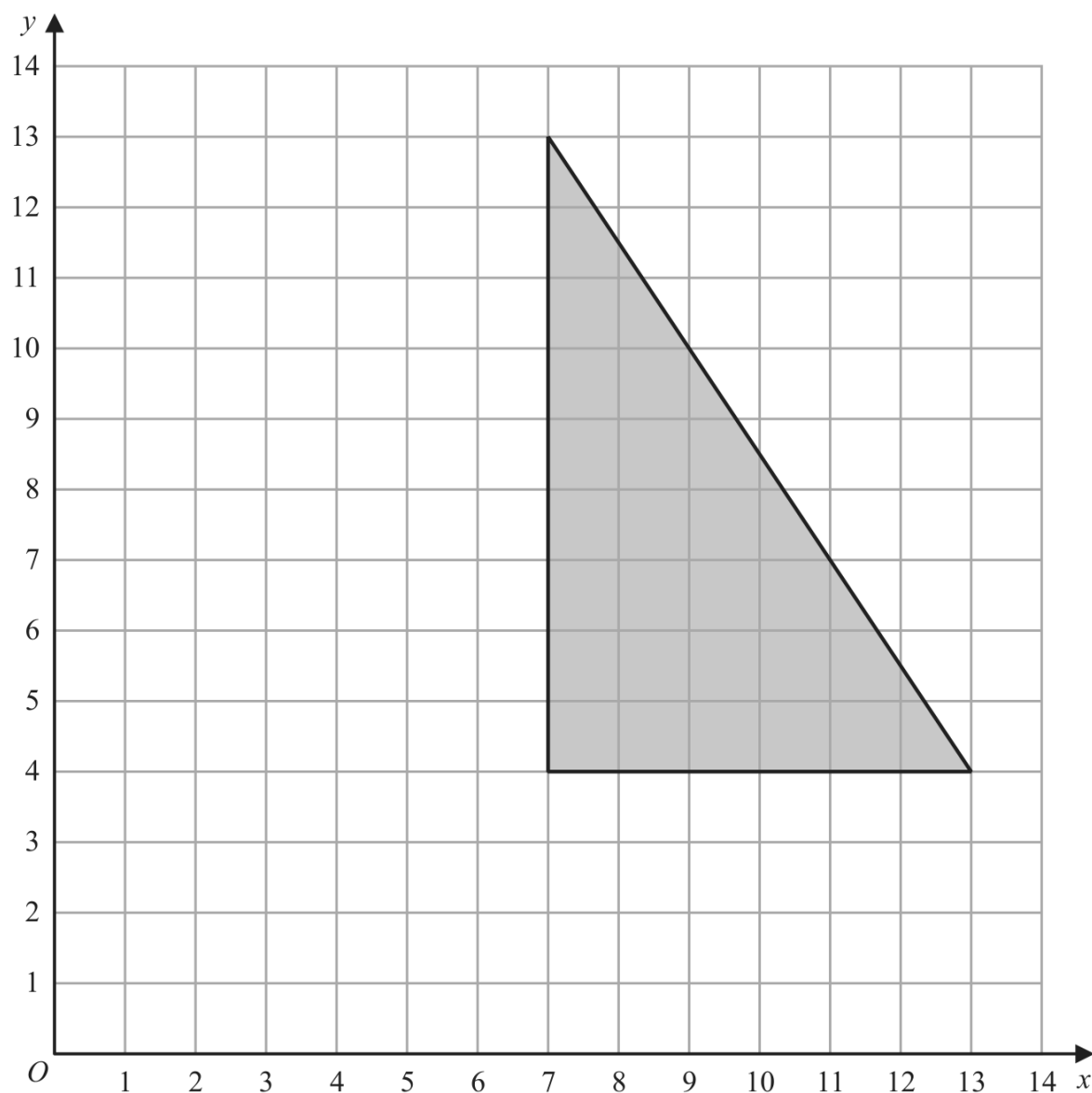
(b) On the grid above, translate triangle A by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$

Label your triangle C

(Total for Question 1 is 4 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q10

CH 5 • TOPIC 2 2 marks**10**

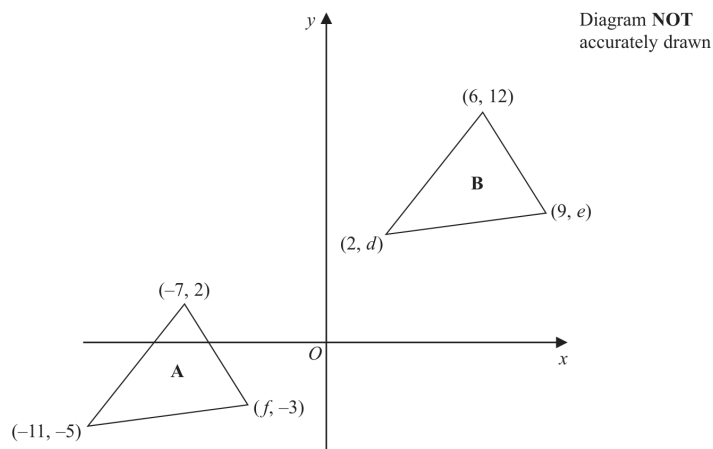
On the grid, enlarge the shaded shape with scale factor $\frac{1}{3}$ and centre (1, 7)

(Total for Question 10 is 2 marks)

Q28 **PAST PAPER** Edexcel 4MA1 • May 2023 • Paper 1H • Q4

CH 5 • TOPIC 2 **8 marks**

- 4 The diagram shows a sketch of triangle **A** and triangle **B** on a coordinate grid.



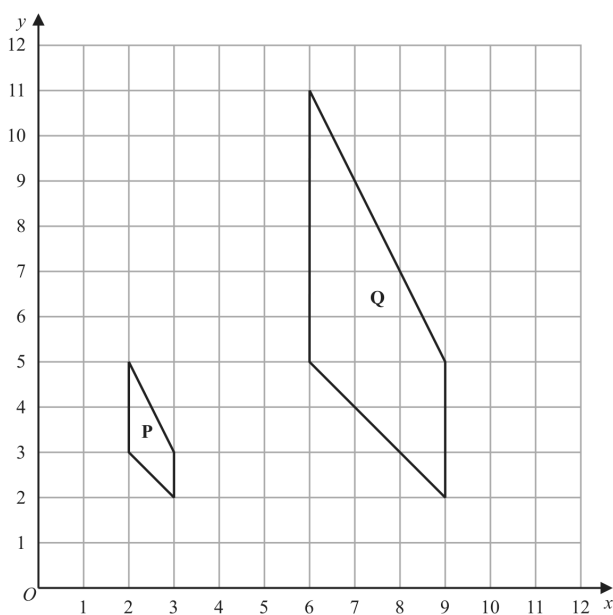
- (a) Given that triangle **A** has been translated to give triangle **B**, work out the value of d , the value of e and the value of f

$d = \dots\dots\dots$

$e = \dots\dots\dots$

$f = \dots\dots\dots$

The diagram shows shape **P** and shape **Q** drawn on a grid.



- (b) Describe fully the single transformation that maps shape **P** onto shape **Q**

- (c) On the grid above, rotate shape **P** 90° clockwise about $(3, 5)$
Label your shape **R**

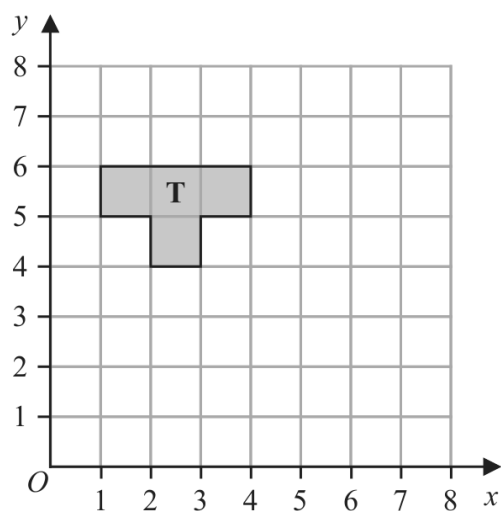
(Total for Question 4 is 8 marks)

Q29**PAST PAPER**

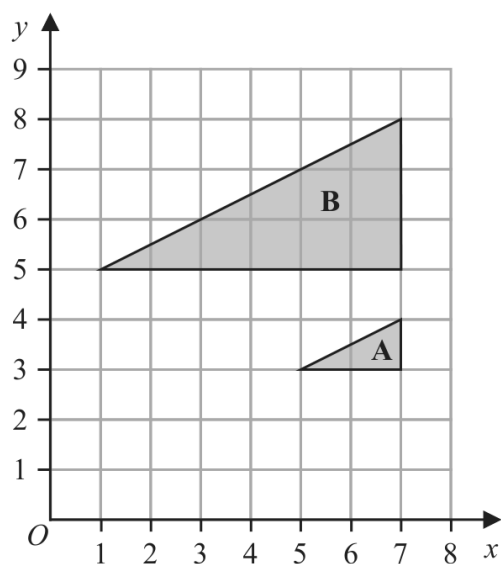
Edexcel 4MA1 • November 2023 • Paper 2H • Q3

CH 5 • TOPIC 2 5 marks

3



- (a) Reflect shape T in the line $y = x$



- (b) Describe fully the single transformation that maps triangle A onto triangle B

(Total for Question 3 is 5 marks)

05 Combined Transformations

Do the first move, then the second move on the new shape. Then compare the start and the end, and describe that one single move.

WHEN YOU SEE → YOU DO

“reflected in $x = -1$ then in $y = 0$ ”

→ two perpendicular mirrors = rotation 180° about where they cross

two parallel mirrors

→ a translation (twice the gap)

“describe the single transformation”

→ one move only — never “reflect then translate”

“is Amy correct?”

→ draw both orders; compare

First move

Second move

Compare start and end

One description

WORKED MODEL • PERPENDICULAR MIRRORS

Shape P is reflected in $x = -1$, then in $y = -2$. Describe the single transformation.

1 the two mirror lines cross at $(-1, -2)$

meeting point

ANSWER Rotation 180° about $(-1, -2)$

× **TRAP** The order matters: reflect then rotate is usually not rotate then reflect.

IDEA 05

Practise: Combined Transformations

My questions first, then selected practice.

Q30 DR ESLAM Combined

CH 5 • TOPIC 2 2 marks

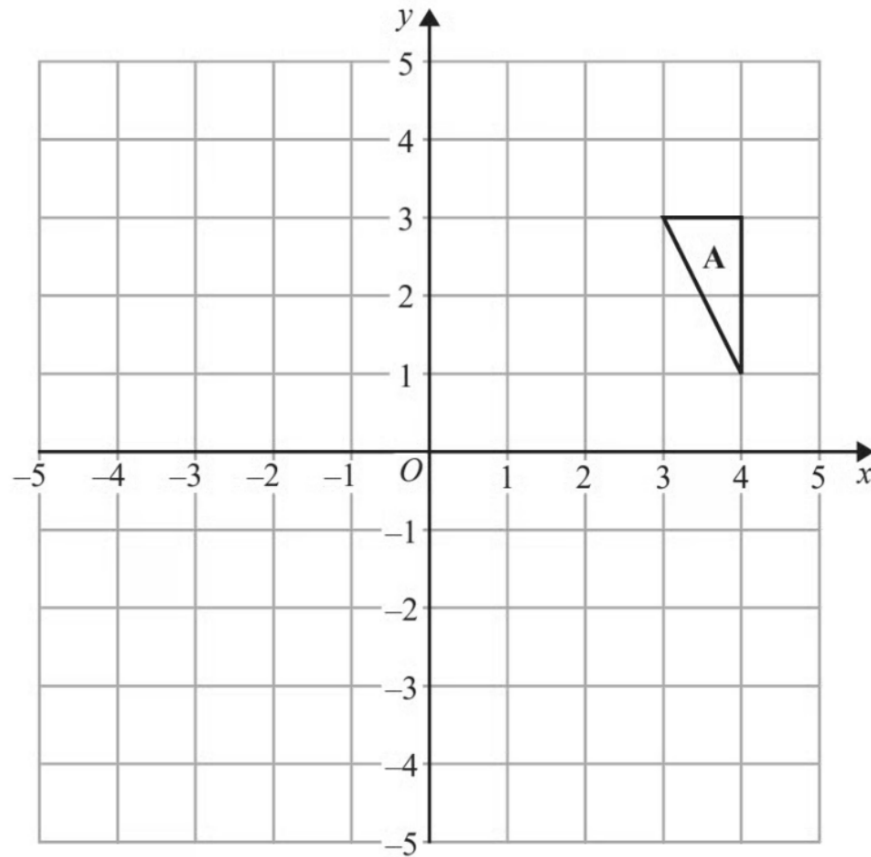
Reflect in $x = 1$, then reflect in $x = 4$. Describe the single transformation.

SELECTED PRACTICE

Q31 PRACTICE Practice • Hard

CH 5 • TOPIC 2 3 marks

2 The diagram shows triangle **A** drawn on a grid.



Kyle reflects triangle **A** in the x -axis to get triangle **B**.

He then reflects triangle **B** in the line $y = x$ to get triangle **C**.

Amy reflects triangle **A** in the line $y = x$ to get triangle **D**.

She is then going to reflect triangle **D** in the x -axis to get triangle **E**.

Amy says that triangle **E** should be in the same position as triangle **C**.

Is Amy correct?

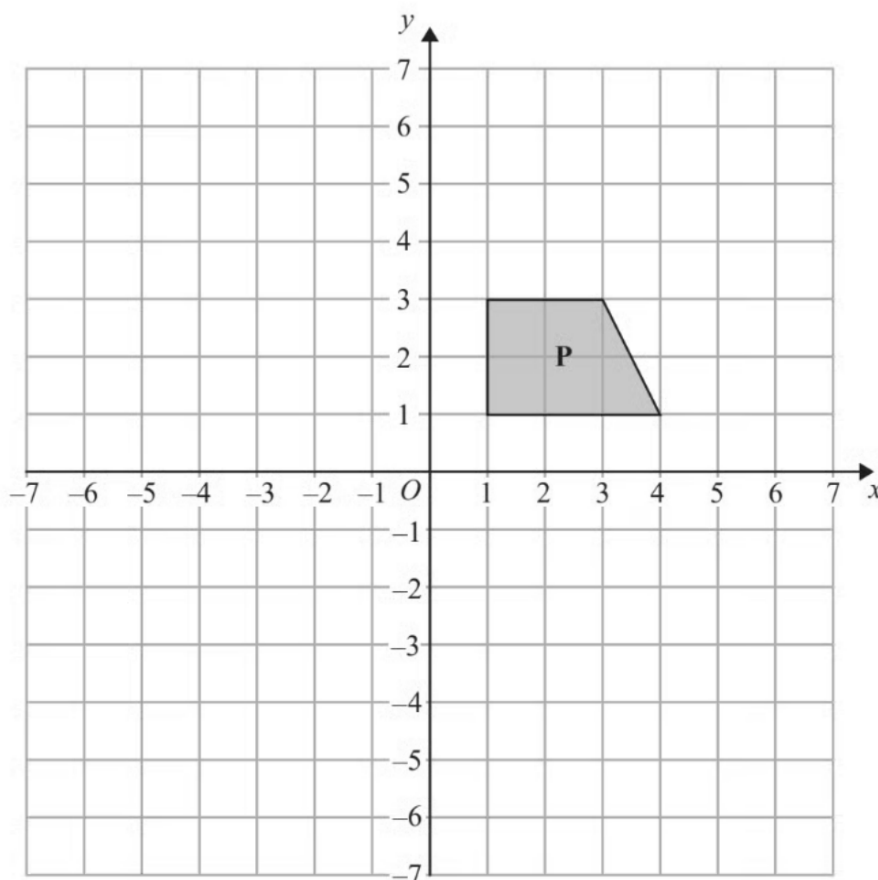
You must show how you get your answer.

(3 marks)

Q32**PRACTICE**

Practice · Hard

CH 5 · TOPIC 2 3 marks

3

Shape **P** is reflected in the line $x = -1$ to give shape **Q**.

Shape **Q** is reflected in the line $y = 0$ to give shape **R**.

Describe fully the **single** transformation that maps shape **P** onto shape **R**.

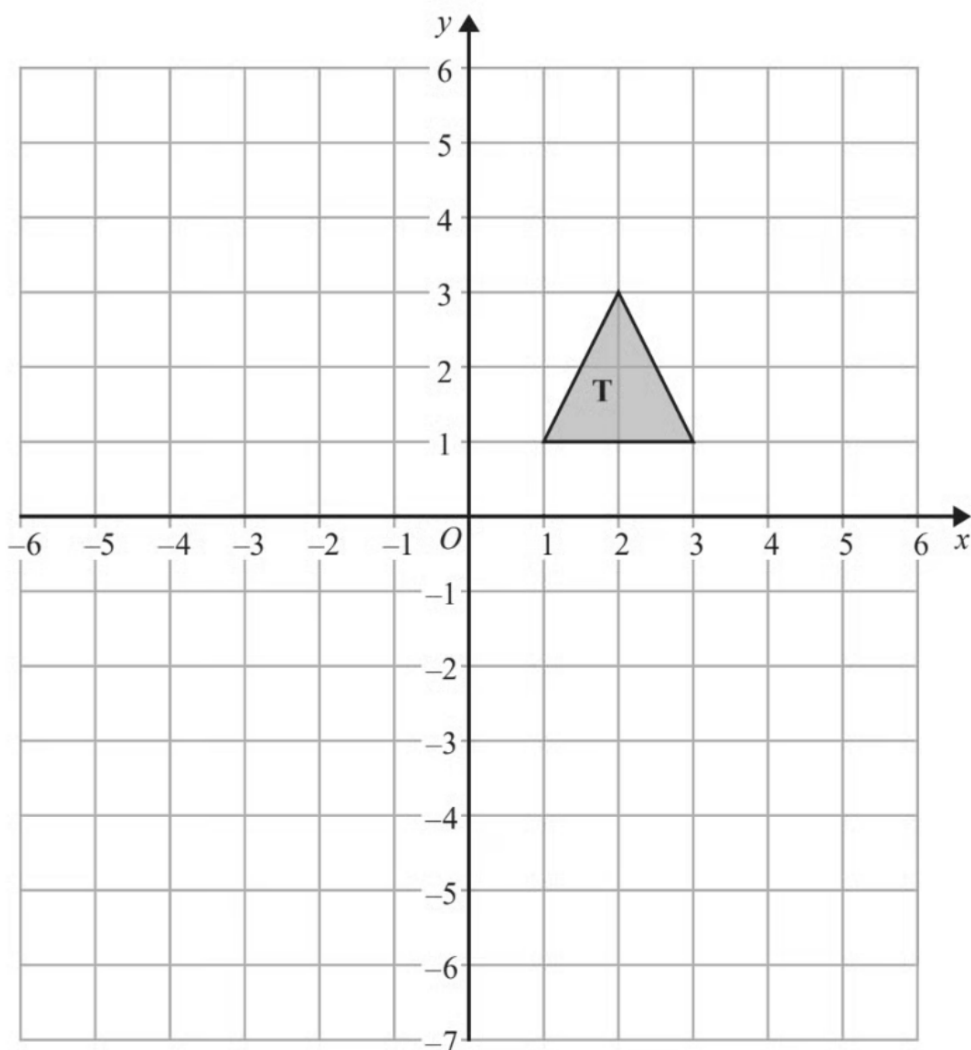
(3 marks)

Q33**PRACTICE**

Practice • Hard

CH 5 • TOPIC 2 2 marks

4



Shape **T** is reflected in the line $x = -1$ to give shape **R**.

Shape **R** is reflected in the line $y = -2$ to give shape **S**.

Describe the **single** transformation that will map shape **T** to shape **S**.

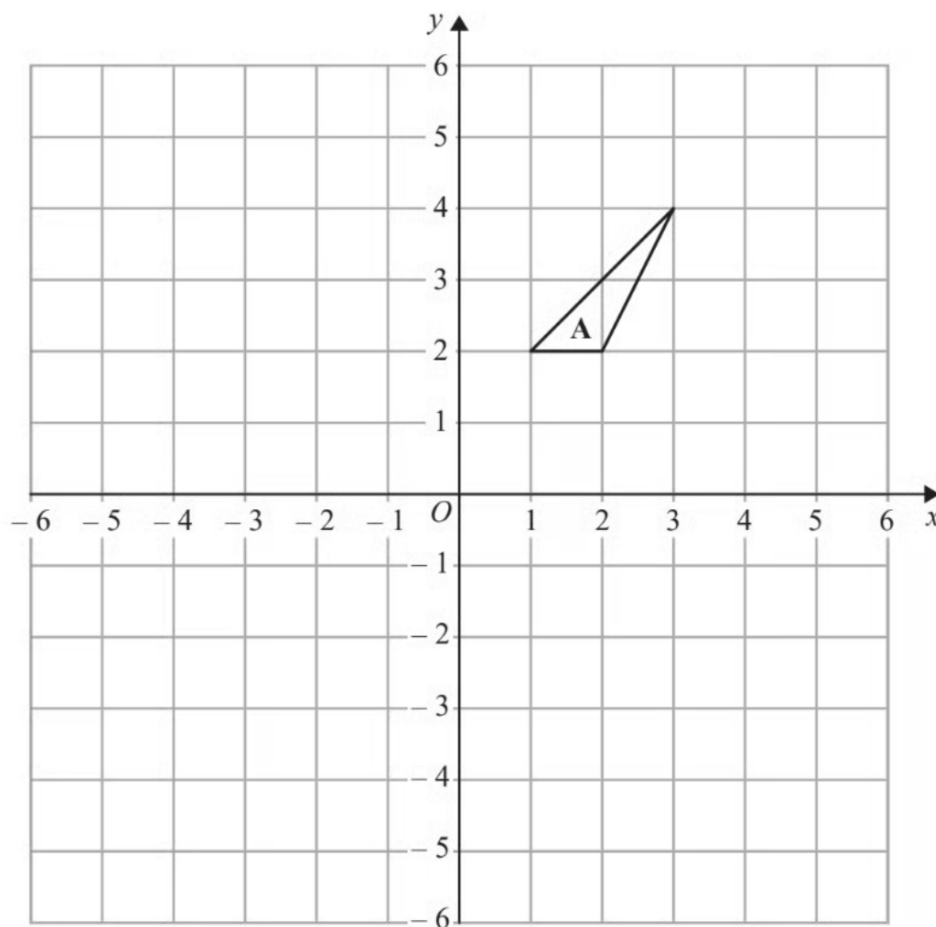
(2 marks)

Q34**PRACTICE**

Practice • Hard

CH 5 • TOPIC 2 3 marks

8



Triangle **A** is rotated 90° clockwise about the point $(0, 1)$ to give triangle **B**. Triangle **B** is translated by the vector $\begin{pmatrix} -3 \\ -1 \end{pmatrix}$ to give triangle **C**. Describe fully the single transformation that maps triangle **A** onto triangle **C**.

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------|---|
| 01 | Translation | Column vector: right/left on top, up/down below. |
| 02 | Reflection | Mirror line equation; $y = x$ swaps the coordinates. |
| 03 | Rotation | Angle, direction, centre — tracing paper. |
| 04 | Enlargement | Scale factor and centre; distances from the centre $\times k$. |
| 05 | Combined | Do both, then describe ONE move. |

BOTTOM LINE Slide $\rightarrow$ vector • flip $\rightarrow$ line • turn $\rightarrow$ angle, direction, centre • resize $\rightarrow$ scale factor, centre.

Answers

Transformations • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • TRANSLATIONS

- Q1** Dr Eslam • Translation $(-2, 6), (1, 6), (-2, 9)$
- Q2** Practice • Easy Translation $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$
- Q3** Practice • Easy $T: (1, -3), (2, -3), (2, -1)$
- Q4** Practice • Easy $p = 5, q = -4$
- Q5** Practice • Medium (a) $(2, -1), (3, -1), (3, 0), (4, 0), (4, 2), (2, 2)$ (b) Rotation 180° , centre $(0, 1)$
- Q6** Edexcel 4MA1 • June 2024 • Paper 2HR • Q2 (a) Translation $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$ (b) C: $(-4, -1), (-4, -2), (-5, -2), (-6, -1)$

IDEA 02 • REFLECTIONS

- Q7** Dr Eslam • Mirror line (a) $(-1, 1)$ (b) $(1, 5)$ (c) $(5, -3)$
- Q8** Practice • Easy (a) $(4, 1), (4, 2), (2, 2), (3, 1)$ (b) Translation $\begin{pmatrix} 4 \\ -1 \end{pmatrix}$
- Q9** Practice • Easy (a) $(1, 5), (4, 5), (4, 7)$ (b) Translation $\begin{pmatrix} -2 \\ -4 \end{pmatrix}$
- Q10** Practice • Medium (a) $(-3, 3), (-3, 4), (-1, 4)$ (b) Reflection in the line $y = x$
- Q11** Practice • Medium $c = -6, d = -1$
- Q12** Edexcel 4MA1 • June 2025 • Paper 2HR • Q5 (a) $(2, 5), (2, 8), (3, 7), (3, 6)$ (b) $(5, 3), (9, 3), (7, 7), (5, 7)$

IDEA 03 • ROTATIONS

- Q13** Dr Eslam • Rotation $(1, -3); (-3, -1)$
- Q14** Practice • Easy (a) A: $(-2, 1), (-4, 1), (-5, 2), (-2, 2)$ (b) B: $(1, -4), (3, -4), (4, -5), (1, -5)$
Rotation 180° , centre $(2, 1)$
- Q15** Practice • Medium (a) $(-1, 1), (-2, 1), (-2, 3)$ (b) Enlargement, scale factor 3, centre $(1, 0)$
- Q16** Practice • Medium Rotation 90° clockwise, centre $(-2, 3)$
- Q17** Practice • Hard
- Q18** Edexcel 4MA1 • January 2021 • Paper 2HR • Q5 (a) Q: $(-6, 6), (-9, 6), (-9, 9), (-3, 9)$ (b) R: $(10, 6), (13, 6), (13, 3), (7, 3)$ (c) Enlargement, scale factor 2, centre $(-3, 3)$
- Q19** Edexcel 4MA1 • June 2025 • Paper 2H • Q2 (a) $(3, 1), (9, 1), (9, 10), (6, 10), (6, 4), (3, 4)$ (b) Rotation 180° , centre $(5, 5)$

IDEA 04 • ENLARGEMENTS

- Q20** Dr Eslam • Enlargement $(-1, -1), (-2, -1), (-1, -3)$
- Q21** Practice • Easy From A: vertices at $(0, 6), (0, 9), (12, 9), (12, 0), (9, 0)$ squares (right, up) — 3 times as far from A
- Q22** Practice • Medium Enlargement, scale factor $\frac{5}{2}$, centre $(0, 0)$
- Q23** Practice • Hard Area $ABC = \frac{1}{2}(4)(3) = 6$; $6 \times (\frac{1}{2})^2 = 1.5 \text{ cm}^2$
- Q24** Practice • Hard (a) BC is not the hypotenuse: $BC^2 = 8^2 - 6^2 = 28$, $BC = 5.29$ (b) $CZ = 10$ squares $= 2\frac{1}{2} \times CR$: scale factor $2\frac{1}{2}$, not $1\frac{1}{2}$
- Q25** Edexcel 4MA1 • January 2020 • Paper 1H • Q8 $(4, 6), (8, 6), (8, 3), (6, 3)$
- Q26** Edexcel 4MA1 • June 2022 • Paper 2HR • Q1 (a) Enlargement, scale factor 3, centre $(3, 0)$
- Q27** Edexcel 4MA1 • May 2023 • Paper 1HR • Q10 (b) C: $(1, 4), (1, 6), (2, 4)$
- Q28** Edexcel 4MA1 • May 2023 • Paper 1H • Q4 $(3, 9), (3, 6), (5, 6)$
- Q29** Edexcel 4MA1 • November 2023 • Paper 2H • Q3 (a) $d = 5, e = 7, f = -4$ (b) Enlargement, scale factor 3, centre $(0, 2)$ (c) R: $(3, 6), (1, 6), (0, 5), (1, 5)$
- (a) $(6, 1), (6, 4), (5, 4), (5, 3), (4, 3), (4, 2), (5, 2), (5, 1)$ (b) Enlargement, scale factor 3, centre $(7, 2)$

IDEA 05 • COMBINED TRANSFORMATIONS

- Q30** Dr Eslam • Combined Translation $\begin{pmatrix} 6 \\ 0 \end{pmatrix}$
- Q31** Practice • Hard C: $(-3, 3), (-3, 4), (-1, 4)$; E: $(3, -3), (3, -4), (1, -4)$. Different — Amy is not correct
- Q32** Practice • Hard Rotation 180° , centre $(-1, 0)$
- Q33** Practice • Hard Rotation 180° , centre $(-1, -2)$
- Q34** Practice • Hard Rotation 90° clockwise, centre $(-2, 2)$